Manufacturing Production Monitoring System

Comprehensive Project Problem Statement

Executive Summary

In today's competitive manufacturing landscape, organizations face significant challenges in tracking production efficiency, machine utilization, and quality control in real-time. Traditional manual data collection methods lead to delayed decision-making, hidden productivity losses, and quality escapes that impact profitability and customer satisfaction.

This project aims to develop an **enterprise-grade Manufacturing Production Monitoring System** that provides real-time visibility into shop floor operations, enables data-driven decision-making, and facilitates continuous improvement through comprehensive analytics. The system will serve as a bridge between shop floor equipment and enterprise business systems, creating a unified digital operations layer.

Background & Problem Context

Current Challenges in Manufacturing Environments:

According to industry research, manufacturers who deploy effective monitoring systems consistently recover **15 to 30 percentage points of hidden production capacity** from existing equipment . However, many plants still operate with:

- **Information silos**: Production, downtime, and performance data remain locked within PLCs, SCADA systems, and disconnected spreadsheets
- **Manual data collection**: Operators spend valuable time recording production counts and downtime reasons on paper forms or basic spreadsheets, leading to data accuracy concerns
- **Delayed visibility**: Supervisors lack live visibility into line performance, delaying corrective actions by hours or shifts
- **Inconsistent quality tracking**: Quality issues surface after production completion, increasing rework costs and customer complaint risk
- **Reactive maintenance**: Equipment failures trigger unplanned downtime rather than condition-based interventions

Business Impact

Without a robust production monitoring system, manufacturing organizations experience:

Impact Area	Consequence
Hidden Capacity Loss	15-30% of productive capacity remains untapped due to undocumented micro-stoppages and inefficiencies
Quality Escapes	Defects detected post-production increase rework costs and potential customer returns
Unplanned Downtime	Calendar-based maintenance misses early warning signs, causing breakdowns
Inaccurate OEE	Manual reporting produces OEE accuracy of only 60-75% vs. 92-98% with automated systems
Audit Risk	Paper-based records require extensive manual retrieval and may lack traceability

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Core System Modules

Module 1: Production Planning

The Production Planning module serves as the bridge between enterprise resource planning (ERP) and shop floor execution. It ensures that production schedules are properly sequenced, resources are allocated efficiently, and actual performance is tracked against planned targets.

1.1 Work Order Management

Feature	Description
Work Order Creation & Scheduling	Create, edit, and schedule production work orders with priority levels, due dates, and routing sequences
Work Order Dispatch	Automatically or manually assign work orders to specific production lines and work centers based on capacity and skill requirements
Real-time Status Tracking	Track work order progress from "Released" → "In Progress" → "Paused" → "Completed" → "Closed"
Order Splitting & Merging	Split large orders across multiple lines or merge small orders for batch efficiency
Priority Sequencing	Dynamic re-prioritization of work orders based on customer urgency, material availability, or machine constraints

1.2 Production Scheduling

Feature	Description

Feature	Description
Gantt Chart Scheduling	Visual timeline showing work orders assigned to each machine/line with drag-and-drop rescheduling capability
Finite Capacity Planning	Schedule optimization that respects actual machine capacity, tooling availability, and operator schedules
Changeover Management	Track and minimize changeover times between product runs with SMED (Single Minute Exchange of Die) analytics
Schedule Adherence Tracking	Compare planned vs. actual production start/completion times with variance reporting
What-If Scenario Analysis	Simulate schedule changes to evaluate impact on delivery dates before committing

1.3 Material Requirements & BOM Integration

Feature	Description
Bill of Materials (BOM) Management	Define product BOMs with component quantities, alternative materials, and consumption rules
Material Availability Check	Automatically verify component availability before releasing work orders
Component Allocation	Reserve and allocate inventory to specific work orders with backflush capability
Kitting Support	Generate picking lists and coordinate kitting operations for complex assemblies

Module 2: Machine Monitoring

The Machine Monitoring module captures real-time production data from equipment through both automated and manual methods, providing visibility into machine status, performance metrics, and loss events.

2.1 Real-time Machine Data Acquisition

Feature	Description
IoT Sensor Integration (Simulated)	Simulated current sensors detect machine running/idle/fault states; for production, supports OPC-UA, Modbus TCP, and MQTT protocols
State Transition Tracking	Automatically record timestamps for state changes: RUN → IDLE → FAULT → SETUP → OFF
Cycle Counting	Accurate production cycle counting with configurable debounce timing

Feature	Description
Parameter Monitoring	Track key machine parameters: temperature, vibration, pressure, speed, energy consumption
Operator Interface	Tablet/terminal interface for operators to declare downtime reasons, log activities, and report quality events

2.2 Data Collection Modes

Based on industry best practices, the system supports multiple data collection architectures :

Mode	Accuracy	Deployment	Best For
IoT Sensor + Cloud (Recommended)	92-98%	48-72 hours	Legacy equipment, mixed fleets, limited IT resources
PLC/OPC-UA Direct Integration	95-100%	2-4 hours per machine	Modern CNC machines, automated lines with digital interfaces
Manual Operator Input	60-75%	Immediate	Low-volume operations, initial awareness phase

2.3 Real-time Dashboard & Visualization

Feature	Description
Live Machine Status Board	Color-coded visualization of all machines showing current state (Running=Green, Idle=Yellow, Fault=Red, Off=Grey)
Performance Gauges	Real-time OEE, Availability, Performance, Quality metrics with target comparisons
Shift Progress Tracking	Countdown timers showing planned vs. actual production against shift targets
Alerts & Notifications	Configurable alerts when machines fall below OEE targets or exceed threshold durations
Role-based Views	Operator-focused dashboards (simple, glanceable) vs. supervisor/manager dashboards (detailed analytics)

2.4 Downtime Tracking & Analysis

Feature	Description
Automated Downtime Detection	System automatically detects and timestamps downtime events based on machine state changes
Reason Categorization	Configurable downtime reason hierarchy (e.g., Mechanical → Bearing Failure, Electrical → Sensor Fault, Operational → Material Shortage)

Feature	Description
Operator Confirmation	Touchscreen interface for operators to confirm/select downtime reasons at event resolution
Micro-stoppage Capture	Detect and log short stops (<2 minutes) that traditional systems miss
Loss Tree Analysis	Drill-down visualization showing top loss categories by shift, product, or machine

Module 3: Defect Tracking

The Defect Tracking module captures quality events in real-time, enabling immediate corrective action and root cause analysis.

3.1 Quality Data Capture

Feature	Description
In-process Inspection	Configurable checkpoints for quality inspections throughout the production process
Defect Classification	Hierarchical defect catalog (e.g., Category: Dimensional → Sub-category: Length → Defect: Over spec by >0.5mm)
Good/Defect Counts	Real-time operator input of good pieces, rework pieces, and scrap pieces at machine or work center
Statistical Process Control (SPC)	Real-time SPC charting for critical quality parameters with control limit alerts
Digital Quality Forms	Replace paper-based inspection forms with guided digital workflows and electronic signatures

3.2 First Pass Yield (FPY) Analytics

Feature	Description
FPY Calculation by Station	Track the percentage of units that pass quality inspection on first attempt without rework
Defect Pareto Analysis	Visual Pareto charts identifying the most frequent defect types for targeted improvement
Yield by Shift/Operator	Comparative yield analysis across shifts and operators to identify training opportunities
Scrap Trending	Trend analysis of scrap rates by product SKU, line, and time period

3.3 Corrective Action Management (CAPA)

Feature	Description
Defect Escalation	Automated escalation when defect rates exceed configurable thresholds
Root Cause Documentation	Structured root cause analysis with 5-Why methodology support
Corrective Action Workflow	Assign, track, and verify corrective actions with due dates and completion verification
Containment Management	Track hold orders, rework orders, and disposition decisions for non-conforming material

Module 4: IoT Simulation

Since the project is for development and demonstration, a comprehensive IoT simulation module will emulate shop floor data without requiring physical sensors or PLC connections.

4.1 Machine Data Emulator

Feature	Description
Configurable Machine Profiles	Define machine types with realistic behavior patterns (automotive assembly, CNC machining, packaging, injection molding)
State Transition Simulation	Emulate realistic machine cycles: Running (45-90 sec) → Occasional Idle (10-30 sec) → Random Faults (2-5 min downtime)
OEE Baseline Generation	Generate data with configurable OEE baseline (e.g., 65% baseline ± random variation)
Multi-line Simulation	Simulate 5-50 machines/lines with independent behavior patterns and shift schedules
Defect Injection	Programmable defect rates that vary by shift, product SKU, or machine condition
Trend Simulation	Simulate gradual performance degradation (e.g., OEE dropping 2% per hour) or improvement (e.g., learning curve effects)

4.2 Simulated Operator Interactions

Feature	Description
Login/Logout Simulation	Simulated operator shift changes with role-based permissions
Downtime Reason Entry	Automatically generate realistic downtime reason selections from operator perspective
Quality Event Entry	Simulate quality checks at configurable intervals with realistic defect data

Feature	Description
Material Replenishment	Simulated material shortage events and resolution

Module 5: Production Analytics

The analytics module transforms raw production data into actionable intelligence through comprehensive reporting, visualization, and predictive insights.

5.1 OEE (Overall Equipment Effectiveness) Analytics

OEE is the gold standard metric, calculated as: **OEE = Availability × Performance × Quality**

Component	Calculation	Target
Availability	(Operating Time / Planned Production Time)	>90%
Performance	(Ideal Cycle Time × Total Parts) / Operating Time	>95%
Quality	Good Parts / Total Parts Produced	>99%
Overall OEE	Availability × Performance × Quality	>85% World Class

Feature	Description
Real-time OEE Calculation	Continuous OEE calculation with drill-down to component metrics
OEE by Machine/Line	Comparative OEE analysis across all equipment
OEE by Shift/Team	Shift performance comparison with trend analysis
OEE by Product/SKU	Identify which products run most efficiently on which equipment
OEE Loss Tree	Visual breakdown of OEE losses: Availability Losses, Performance Losses, Quality Losses

5.2 Production Performance Dashboards

Feature	Description
Tiered Dashboards	Operator view (real-time OEE, target vs. actual), Supervisor view (shift performance, top losses), Manager view (trends, cross-line comparison)
Hourly KPI Tracking	Track performance by hour to identify intra-shift issues
Throughput Analysis	Production output trends by hour, shift, day, week, month

Feature	Description
Bottleneck Identification	Automatically identify constraint operations limiting overall line throughput
Cycle Time Analysis	Track actual vs. standard cycle times with statistical distribution

5.3 Reporting Engine

Feature	Description
Automated Report Generation	Scheduled email delivery of shift summaries, daily production reports, weekly OEE reviews
Export Capabilities	CSV, Excel, PDF export with configurable date ranges and filters
Custom Report Builder	Drag-and-drop report designer for users to create custom analytics
Executive Summary	High-level dashboard for plant management with key metrics at-a-glance
Audit Trail Reports	Complete traceability of all production events, quality checks, and user actions with timestamp and user attribution

5.4 Trend Analysis & Predictive Insights

Feature	Description
Performance Trending	30/60/90 day trend analysis for OEE, FPY, Downtime by category
MTBF/MTTR Tracking	Mean Time Between Failures and Mean Time To Repair analytics for maintenance planning
Seasonality Detection	Identify patterns by day of week, time of day, or production cycle
Anomaly Detection	Statistical identification of unusual performance patterns requiring investigation
Predictive Alerts (ML Foundation)	Alert when predictive models indicate increased failure risk based on parameter deviations

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System Users, Roles, and Responsibilities

Based on MES access control best practices, the system implements **Role-Based Access Control (RBAC)** with segregation of duties (SoD) to ensure data integrity and compliance .

User Role Matrix

Role	Primary Location	Access Level	Key Responsibilities
Operator	Shop Floor	Operate	Start/stop production, log downtime reasons, record quality checks, report defects
Production Supervisor	Shop Floor/Office	Supervise	Monitor real-time performance, assign work orders, respond to alerts, approve quality exceptions
Quality Inspector	Shop Floor/Lab	Quality	Perform inspections, log defects, approve/reject batches, initiate CAPA
Maintenance Technician	Shop Floor	Maintenance	Acknowledge machine faults, log repair activities, track MTTR, schedule preventive maintenance
Production Planner	Office	Plan	Create work orders, schedule production, adjust priorities, analyze capacity
Plant Manager	Office	Manage	View KPIs, analyze trends, review OEE reports, approve CAPA
System Administrator	IT	Administer	User management, role configuration, system configuration, audit log review

Detailed Role Definitions

Operator

Count per plant: 5-50 (varies by shift and number of machines)

Access locations: Shop floor terminals/tablets, minimal web access

Responsibility	Description
Shift Login/Logout	Log into assigned workstation at shift start, select current work order
Production Count Entry	Record good pieces produced (automatically from IoT simulation)
Downtime Reason Selection	When machine stops, select reason from categorized hierarchy within 30 seconds of resolution
Quality Check Reporting	Perform and record in-process inspections at defined intervals
Defect Logging	Record defective pieces with defect type and quantity
Material Replenishment Request	Trigger material request when supplies run low
End-of-shift Handover	Complete shift summary and notes for next shift

Segregation of Duties Constraints: Cannot approve quality exceptions, cannot modify production schedules, cannot delete completed transactions

Production Supervisor

Count per plant: 2-5 (one per shift, plus day shift supervisor)

Access locations: Shop floor + office workstation + mobile

Responsibility	Description
Real-time Performance Monitoring	Monitor live OEE, machine status, and production progress across assigned area
Alert Response	Respond to system alerts when OEE drops below thresholds or machines exceed downtime limits
Work Order Management	Release, pause, or reprioritize work orders based on real-time conditions
Downtime Validation	Review and approve operator-entered downtime reasons for accuracy
Quality Exception Approval	Approve rework disposition or scrap authorization
Shift Performance Review	Lead shift handover meetings using system-generated performance reports
Team Coaching	Use analytics to identify operator training needs and improvement opportunities

Quality Inspector

Count per plant: 1-3

Access locations: Quality lab + shop floor + office

Responsibility	Description
In-process Audits	Perform random quality audits and record findings in system
Defect Verification	Verify operator-reported defects and update classifications as needed
Non-conformance Management	Create and manage non-conformance reports (NCRs)
Corrective Action Initiation	Initiate CAPA records for significant quality issues
Batch Disposition	Approve batch release, rework, or scrap decisions
SPC Monitoring	Review SPC charts and investigate out-of-control conditions

Maintenance Technician

Count per plant: 2-6

Access locations: Shop floor + mobile

Responsibility	Description
Fault Acknowledgment	Claim machine fault events and provide estimated repair time
Repair Logging	Record actions taken, parts replaced, and time spent
Preventive Maintenance Execution	Complete scheduled PM tasks and document findings
MTTR Optimization	Review repair history to identify opportunities for faster recovery
Spare Parts Request	Request parts through integrated workflow

Production Planner

Count per plant: 1-2

Access locations: Office workstation

Responsibility	Description
Master Scheduling	Create master production schedule based on demand forecasts
Work Order Creation	Generate detailed work orders with BOMs, routings, and due dates
Capacity Planning	Analyze capacity constraints and level-load production
Changeover Optimization	Sequence production to minimize changeover time
Schedule Adherence Reporting	Track and report on schedule attainment

Plant Manager

Count per plant: 1

Access locations: Office + mobile

Responsibility	Description
KPI Review	Review plant-wide OEE, FPY, throughput, and cost metrics
Trend Analysis	Analyze performance trends to guide strategic improvement
CAPA Approval	Approve major corrective actions and verify effectiveness
Budget & Target Setting	Set production targets and approve improvement initiative budgets
Executive Reporting	Generate reports for senior leadership and board reviews

System Administrator

Count per plant: 1 (may support multiple plants)

Access locations: IT office (remote possible)

Responsibility	Description
User Access Management	Create, modify, and disable user accounts; conduct quarterly access reviews
Role Configuration	Define roles and permission sets; enforce segregation of duties
System Configuration	Configure machines, work centers, shift calendars, and downtime reason hierarchies
Integration Management	Manage API integrations and data synchronization
Audit Log Review	Review system audit trails for compliance and security incidents
Backup & Recovery	Ensure regular backups and test recovery procedures

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Technical Architecture

Recommended Technology Stack

Based on the requirement for PostgreSQL, ASP.NET Core Web API, and Angular:

Layer	Technology	Purpose
Database	PostgreSQL	Primary data storage with JSON support for flexible machine configurations
Backend API	ASP.NET Core 8/9 Web API	RESTful API with Swagger documentation, JWT authentication
Frontend	Angular 17+	Single-page application with responsive design, real-time updates via SignalR
Real-time Communication	SignalR	Live dashboard updates, alerts, and machine state changes
Authentication	JWT + ASP.NET Core Identity	Secure authentication and role-based authorization
Reporting	Angular + jsPDF / ExcelJS	Native report generation and export
IoT Simulation	Background Service (IHostedService)	Simulated machine data generation and state management
Caching	Redis (optional)	Performance optimization for frequently accessed data

Layer	Technology	Purpose
Logging	Serilog + PostgreSQL	Structured logging with queryable log store

Database Schema Highlights

Entity	Description
`Machines`	Machine master data: name, type, ideal cycle time, OEE target
`WorkOrders`	Production orders: order number, product, quantity, status, dates
`ProductionEvents`	Time-stamped records of machine state changes, counts, and events
`DowntimeRecords`	Downtime events with start/end times, reasons, categories
`QualityChecks`	Inspection records: pass/fail, measurements, defect codes
`Defects`	Defect entries linked to production events
`Users/Roles`	ASP.NET Core Identity tables + custom role definitions
`AuditLogs`	Complete audit trail of all data changes with user attribution

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Key Performance Indicators (KPIs)

The system will track and report the following manufacturing KPIs:

KPI	Formula	Target	Update Frequency
OEE	Availability × Performance × Quality	>85%	Real-time
Availability	Operating Time / Planned Production Time	>90%	Real-time
Performance	(Ideal Cycle Time × Total Parts) / Operating Time	>95%	Real-time
Quality / FPY	Good Parts / Total Parts	>99%	Real-time
MTBF	Total Operating Time / Number of Breakdowns	>500 hours	Daily
MTTR	Total Repair Time / Number of Breakdowns	<30 minutes	Daily
Schedule Adherence	(Actual Completion Time - Planned Time) / Planned Time	±5%	Per order
Scrap Rate	Scrap Quantity / Total Produced Quantity	<1%	Per shift

KPI	Formula	Target	Update Frequency
Changeover Time	Time between last good part and first good part of next run	<15 minutes	Per changeover

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Implementation Phases

Phase 1: Foundation (Weeks 1-4)

- Database schema design and migration
- ASP.NET Core Web API with JWT authentication
- User management and role-based access control
- Machine master data management
- IoT simulation engine

Phase 2: Core Production Tracking (Weeks 5-8)

- Work order management
- Real-time machine monitoring dashboard (Angular + SignalR)
- Production event recording
- Downtime tracking with reason codes

Phase 3: Quality Management (Weeks 9-12)

- Defect tracking module
- Quality check management
- First Pass Yield analytics
- Digital quality forms

Phase 4: Analytics & Reporting (Weeks 13-16)

- OEE calculation and dashboards
- Production analytics and trend analysis
- Report generation and export
- Executive dashboard

Phase 5: Advanced Features (Weeks 17-20)

- CAPA workflows

- Predictive analytics foundation
- Mobile-responsive design
- System administration tools

Success Criteria

The project will be considered successful when:

1. **Real-time Visibility:** All machines display live status with <5 second latency
2. **Automated Data Capture:** OEE accuracy of 92% or higher compared to manual audit
3. **User Adoption:** All operators, supervisors, and managers actively using the system daily
4. **Actionable Analytics:** Users can identify top 3 loss reasons within 2 clicks from dashboard
5. **Data Integrity:** Complete audit trail exists for all production events
6. **Integration Ready:** API documentation complete for ERP integration
7. **Scalable Architecture:** System supports simulation of up to 100 machines simultaneously

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References

This problem statement draws from industry best practices documented in manufacturing monitoring literature, including:

- Real-time OEE monitoring and IoT sensor integration patterns
- Digital factory transformation and loss elimination strategies
- MES access control and role-based security frameworks
- Shop floor data visualization and operator engagement techniques
- Manufacturing analytics and KPI frameworks